MATHEMATICS II FOR DATA SCIENCE
GA-2 SOLUTIONS MAY 2024

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1. Question

In a particular year, the profit (in lakhs of ₹) of Star Fish company is given by the polynomial $P(x) = ax^2 + bx + c$ where x denotes the number of months since the beginning of the year (i.e., $x = 1$ denotes January, $x = 2$ denotes February, and so on). In January and February the company made a loss of ₹45(in lakhs), and ₹19(in lakhs) respectively, and in March the company made a profit of ₹3(in lakhs). Let the loss be represented by negative of profit. Choose the correct set of options based on the given information.

- ☐ The maximum profit will be in the month of May.
- ☐ The maximum profit will be in the month of August.
- ☐ The maximum monthly profit amount is ₹53 lakh.
- ☐ The maximum monthly profit amount is ₹35 lakh.

1.1. Answers:

- The maximum profit will be in the month of August
- The maximum profit amount is ₹53 lakh.

1.2. Solution

With the given information, let us find out $a,\ b\ and\ c$.
When $x = 1$, $P(x) = -45$. When $x = 2$, $P(x) = -19$. When $x = 3$, $P(x) = 3$.

When $x = 1$,

$$a(1)^2 + b(1) + c = -45$$
$$a + b + c = -45 \quad (1)$$

When $x = 2$,

$$a(2)^2 + b(2) + c = -19$$
$$4a + 2b + c = -19 \quad (2)$$

When $x = 3$,

$$a(3)^2 + b(3) + c = 3$$
$$9a + 3b + c = 3 \quad (3)$$

Matrix form $(Ax = b)$ for equations $(1), (2)\ and\ (3):$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \end{bmatrix}, \; x = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \; b = \begin{bmatrix} -45 \\ -19 \\ 3 \end{bmatrix}$$

$$Ax = b \Longrightarrow x = A^{-1}b$$

$A^{-1} = \dfrac{1}{det(A)} adj(A),$

$$\begin{aligned} det(A) &= 1(2-3) - 1(4-9) + 1(12-18) \\ det(A) &= -1 + 5 - 6 \\ det(A) &= -2 \end{aligned}$$

$adj(A) = C_A^T$,. where C_A is the $cofactor$ matrix of A,

$$adj(A) = \begin{bmatrix} (2-3) & -(4-9) & (12-18) \\ -(1-3) & (1-9) & -(3-9) \\ (1-2) & -(1-4) & (2-4) \end{bmatrix}^T$$

$$adj(A) = \begin{bmatrix} -1 & 5 & -6 \\ 2 & -8 & 6 \\ -1 & 3 & -2 \end{bmatrix}^T$$

$$adj(A) = \begin{bmatrix} -1 & 2 & -1 \\ 5 & -8 & 3 \\ -6 & 6 & -2 \end{bmatrix}$$

$x = A^{-1}b \Longrightarrow x = \dfrac{1}{det(A)} adj(A) \times b,$

$$x = \frac{1}{-2} \begin{bmatrix} -1 & 2 & -1 \\ 5 & -8 & 3 \\ -6 & 6 & -2 \end{bmatrix} \times \begin{bmatrix} -45 \\ -19 \\ 3 \end{bmatrix}$$

$$x = \frac{-1}{2} \begin{bmatrix} 45 - 38 - 3 \\ -225 + 152 + 9 \\ 270 - 114 - 6 \end{bmatrix}$$

$$x = \frac{-1}{2} \begin{bmatrix} 4 \\ -64 \\ 150 \end{bmatrix}$$

$$x = \begin{bmatrix} -2 \\ 32 \\ -75 \end{bmatrix}$$

$P(x) = -2x^2 + 32x - 75$

The co-efficient of x^2 is lesser than 0. Hence, P(x) has global maximum. By equating the derivative of P(x) to 0, we get :

$$\begin{aligned} P^{'}(x) &= 0 \\ -4x + 32 &= 0 \\ x &= 8 \end{aligned}$$

$P(x)$ is maximum when x=8. So, the maximum profit is $P(8)$.

$$\begin{aligned} P(8) &= -2(8)^2 + 32(8) - 75 \\ &= -128 + 256 - 75 \\ &= 53 \end{aligned}$$

Thus, the maximum prorfit is ₹$53\ Lakhs$. The maximum will be in the month of August.

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2. Question

If A be a 3×4 matrix and b be a 3×1 matrix, then choose the set of correct options.

1. If $(A \mid b)$ be the augmented matrix and $(A' \mid b')$ be the matrix obtained from $(A \mid b)$ after a finite number of elementary row operations then the system $Ax = b$ and the system $A'x = b'$ have the same set of solutions.
2. If $(A' \mid b')$ is the reduced row echelon form of $(A \mid b)$ then the system $A'x = b'$ has at least one solution.
3. If $(A' \mid b')$ is the reduced row echelon form of $(A \mid b)$, then A' is also in reduced row echelon form.
4. If $(A' \mid b')$ is the reduced row echelon form of $(A \mid b)$, and there is no row such that the only non zero entry lies in the last column of $(A' \mid b')$ then the system $Ax = b$ has at least one solution.

2.1. Answers :

- If $(A \mid b)$ be the augmented matrix and $(A' \mid b')$ be the matrix obtained from $(A \mid b)$ after a finite number of elementary row operations then the system $Ax = b$ and the system $A'x = b'$ have the same set of solutions.
- If $(A' \mid b')$ is the reduced row echelon form of $(A \mid b)$, then A' is also in reduced row echelon form.
- If $(A' \mid b')$ is the reduced row echelon form of $(A \mid b)$, and there is no row such that the only non zero entry lies in the last column of $(A' \mid b')$ then the system $Ax = b$ has at least one solution.

2.2. Solution

- Elementary row operations do not change the solution(s) for a system of linear equations. Hence, **option 1** is correct
- If all the elements in the last row of A' are zeros and the corresponding element in the last row of b' is not zero, then $A'x = b'$ does not have any solution. Hence, **option 2** is incorrect.
- If $(A'|b')$ is in reduced row echelon form, then A' will also be in reduced row echelon form. Hence, **option 3** is correct.
- The last column of $(A'|b')$ is b'. If there is no row of the form $[0\ 0\ \cdots\ 0 \mid p]$, where $p \neq 0$, then there is atleast one solution. Hence **option 4** is correct.

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3. Question

Choose the set of correct options

1. If the sum of all the elements of each row of a matrix A is 0, then A is not invertible.
2. If E is a matrix of order 3×3 obtained from the identity matrix by a finite number of elementary row operations then E is invertible.
3. Any system of linear equations has at least one solution.
4. If A is a matrix of order 3×3 and $det(A) = 3$ then $det(Adj(A)) = 3$.
5. If A is a matrix of order 3×3 and $det(A) = 3$ then $det(Adj(A)) = 9$.

3.1. Answers :

- If the sum of all the elements of each row of a matrix A is 0, then A is not invertible.
- If E is a matrix of order 3×3 obtained from the identity matrix by a finite number of elementary row operations then E is invertible.
- If A is a matrix of order 3×3 and $det(A) = 3$ then $det(Adj(A)) = 9$.

3.2. Solution

- Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$.

  Since the sum of all the elemens of every row is 0, we get the following equations:

$$\begin{aligned} a+b+c &= 0 \\ c &= -a-b \quad (1) \\ d+e+f &= 0 \\ f &= -d-e \quad (2) \\ g+h+i &= 0 \\ i &= -g-h \quad (3) \end{aligned}$$

  By $(1), (2)\ and\ (3)$, we get:

$$A = \begin{bmatrix} a & b & -a-b \\ d & e & -d-e \\ g & h & -g-h \end{bmatrix}$$

  Apply column operations on A

$$\begin{bmatrix} a & b & -a-b \\ d & e & -d-e \\ g & h & -g-h \end{bmatrix} \xrightarrow{\text{C3+C1}} \begin{bmatrix} a & b & -b \\ d & e & -e \\ g & h & -h \end{bmatrix} \xrightarrow{\text{C3+C2}} \begin{bmatrix} a & b & 0 \\ d & e & 0 \\ g & h & 0 \end{bmatrix}$$

  All the entries of the third column are 0. Thereby, $det(A) = 0$. As a result, A is not invertible.

  Hence **option 1** is correct.

- Of the three row operations, interchanging two rows and multiplying a row by a constant would

change the determinant of the resultant matrix by a multiple. Interchanging two rows would multiply the determinant of the new matrix by -1, Scaling a row by a constant c, would multiply the determinant by c. Note, $c \neq 0$.

The determinant of identity matrix is not equal to 0, which would mean that $det(E) \neq 0$. E is invertible. Hence, **option 2** is correct.

- All system of linear equations need not have atleast one solution. For example, $x + y = 3$ and $x + y = 4$ do not have any solution. Hence, **option 3** is incorrect.
- For option 4 and 5 :

  Properties to remember:

  - $det\left(A^{-1}\right) \times det(A) = 1$
  - $det(kA) = k^n det(A)$, where n is the order of the square matrix A and $k \in \mathbb{R}$

$$
\begin{aligned}
A^{-1} &= \frac{1}{det(A)} adj(A) \\
det\left(A^{-1}\right) &= det\left(\frac{1}{det(A)} adj(A)\right) \\
det\left(A^{-1}\right) &= \frac{1}{det(A)^3} det(adj(A)) \\
det(A)^2 &= det(adj(A))
\end{aligned}
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- Given that $det(A) = 3$, we get $det(adj(A)) = 9$. Hence, **option 4** is incorrect and **option 5** is correct.

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4. Question

Ramya bought 1 comic book, 2 horror books, and 1 novel from a bookshop which cost her ₹1000. Romy bought 2 comic books, 5 horror books, and 1 novel which cost him ₹2000. Farjana bought 4 comic books, 5 horror books, and c novels from a shop which cost her ₹d. If x_1 , x_2, and x_3 represent the price of each comic book, horror book, and novel, respectively, then choose the set of correct options.

- [ ] The matrix representation to find x_1, x_2 and x_3 is $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 1 \\ 4 & 5 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1000 \\ 2000 \\ d \end{bmatrix}$
- [ ] The matrix representation to find x_1, x_2 and x_3 is $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 5 \\ 1 & 1 & c \end{bmatrix} = \begin{bmatrix} 1000 & 2000 & d \end{bmatrix}$
- [ ] The matrix representation to find x_1, x_2 and x_3 is $\begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 5 \\ 1 & 1 & c \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1000 & 2000 & d \end{bmatrix}$
- [ ] If Farjana tries to find $x_1, x_2,$ and x_3 using appropriate matrix representation by taking $c = 2$ and $d = 4000$, then the price of each comic book that she thus arrives at, will not be unique.
- [ ] If $c = 7$ and $d = 4000$, then the price of each comic book cannot be determined from this data.
- [ ] If $c = 7$ and $d = 3000$, then the shopkeeper has made a mistake.
- [ ] If $c = 2$ and $d = 3000$, then the price of each comic book can be determined from the data.

4.1. Answers :

- The matrix representation to find x_1, x_2 *and* x_3 is $\begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 1 \\ 4 & 5 & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1000 \\ 2000 \\ d \end{bmatrix}$
- The matrix representation to find x_1, x_2 *and* x_3 is

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 5 \\ 1 & 1 & c \end{bmatrix} = \begin{bmatrix} 1000 & 2000 & d \end{bmatrix}$$

- If $c = 7$ and $d = 4000$, then the price of each comic book cannot be determined from this data.
- If $c = 7$ and $d = 3000$, then the shopkeeper has made a mistake.
- If $c = 2$ and $d = 3000$, then the price of each comic book can be determined from this data.

4.2. Solution

The following are the system of linear equations for the given problem:

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 1000 \\ 2x_1 + 5x_2 + x_3 &= 2000 \\ 4x_1 + 5x_2 + cx_3 &= d \end{aligned}$$

Matrix representation $Ax = b$ for the above system of linear equations is:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 1 \\ 4 & 5 & c \end{bmatrix}, \; x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \; b = \begin{bmatrix} 1000 \\ 2000 \\ d \end{bmatrix}$$

Hence, **option 1** is correct.

$Ax = b \implies (Ax)^T = b^T \implies x^T A^T = b^T$. This can be represented as:

$$\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 4 \\ 2 & 5 & 5 \\ 1 & 1 & c \end{bmatrix} = \begin{bmatrix} 1000 & 2000 & d \end{bmatrix}$$

Hence, **option 2** is correct
Option 3 is incorrect. Multiplication of 3×3 matrix and 1×3 matrix is not possible.
Augment A and b,

$$(A|b) = \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1000 \\ 2 & 5 & 1 & 2000 \\ 4 & 5 & c & d \end{array}\right]$$

Appying elementary row operations to $(A|B)$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1000 \\ 2 & 5 & 1 & 2000 \\ 4 & 5 & c & d \end{array}\right] \xrightarrow{R2-(2)R1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1000 \\ 0 & 1 & -1 & 0 \\ 4 & 5 & c & d \end{array}\right] \xrightarrow{R3-(4)R1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1000 \\ 0 & 1 & -1 & 0 \\ 0 & -3 & c-4 & d-4000 \end{array}\right]$$

$$\xrightarrow{R3+(3)R2}\left[\begin{array}{ccc|c}1&2&1&1000\\0&1&-1&0\\0&0&c-7&d-4000\end{array}\right]\xrightarrow[c-7]{R3}\left[\begin{array}{ccc|c}1&2&1&1000\\0&1&-1&0\\0&0&1&\frac{d-4000}{c-7}\end{array}\right]\xrightarrow{R2+R3}\left[\begin{array}{ccc|c}1&2&1&1000\\0&1&0&\frac{d-4000}{c-7}\\0&0&1&\frac{d-4000}{c-7}\end{array}\right]$$

$$\xrightarrow{R1-(2)R2}\left[\begin{array}{ccc|c}1&0&1&1000-2\left(\frac{d-4000}{c-7}\right)\\0&1&0&\frac{d-4000}{c-7}\\0&0&1&\frac{d-4000}{c-7}\end{array}\right]\xrightarrow{R1-R3}\left[\begin{array}{ccc|c}1&0&0&1000-2\left(\frac{d-4000}{c-7}\right)-\left(\frac{d-4000}{c-7}\right)\\0&1&0&\frac{d-4000}{c-7}\\0&0&1&\frac{d-4000}{c-7}\end{array}\right]$$

Now, the system of linear equations are:

$$\begin{aligned}x_1&=1000-3\left(\frac{d-4000}{c-7}\right)\\x_2&=\frac{d-4000}{c-7}\\x3&=\frac{d-4000}{c-7}\end{aligned}$$

As we can see, if $c=7$, then the denominator will become 0. This would mean that the price cannot be determined. If $c=7$, the shop keeper has made some mistake, irrespective of the value of d. Hence, **option 5** and **option 6** are correct.

If $c=2\ and\ d=3000$, then $\frac{d-4000}{c-7}=200$. Now, $x_1=400,\ x_2=200,\ x_3=200$. Hence, **option 7** is correct.

If $c=2\ and\ d=4000$, then $\frac{d-4000}{c-7}=0$. Now, $x_1=1000,\ x_2=0,\ x_3=0$. The solution is unique. Hence, **option 4** is incorrect.

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5. Question

How many solutions does the following system have? Enter 1 if there is only one solution, 0 if there are no solutions and -1 if there are infinitely many solutions. The coefficient matrix A and the vector b are given below:

$$A=\begin{bmatrix}1&0&-1\\1&1&-1\\2&1&-2\end{bmatrix},\ b=\begin{bmatrix}1\\3\\4\end{bmatrix}$$

5.1. Answer : –1

5.2. Solution

$(A|b) = \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 1 & 1 & -1 & 3 \\ 2 & 1 & -2 & 4 \end{array}\right]$. Applying elementary row operations to $(A|b)$ below:

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 1 & 1 & -1 & 3 \\ 2 & 1 & -2 & 4 \end{array}\right] \xrightarrow{R2-R1} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 2 \\ 2 & 1 & -2 & 4 \end{array}\right] \xrightarrow{R3-2R2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 2 \end{array}\right] \xrightarrow{R3-R2} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Now, the new matrix is in reduced row echelon form. As we can see, there is no row in the new matrix which is the form $\left[\begin{array}{ccc|c} 0 & 0 & 0 & c \end{array}\right]$, $where\ c \neq 0\ and\ c \in \mathbb{R}$. Therefore, the system of linear equations is consistent.
The third column of the new matrix has no leading entry. Hence, x_3 is an independent variable.
If a system linear equations is consistent and it has at least one independent variable then the system of linear equations have infinitely many solutions. Hence, the answer is -1.

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6. Question

Let the reduced row echelon form of a matrix A be $\begin{bmatrix} 1 & 0 & 0 & \frac{-1}{2} \\ 0 & 1 & 0 & \frac{-1}{6} \\ 0 & 0 & 1 & \frac{1}{6} \end{bmatrix}$. The first, second, third and fourth columns of A are $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} a \\ b \\ c \end{bmatrix}\ and\ \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$ respectively. The value of $a+b+c$ is

6.1. Answer : 0

6.2. Solution

Let us apply elementary row operations to the original matrix A.

$$\begin{bmatrix} 1 & 3 & a & -1 \\ 0 & 2 & b & 0 \\ -1 & 1 & c & 0 \end{bmatrix} \xrightarrow{R3+R1} \begin{bmatrix} 1 & 3 & a & -1 \\ 0 & 2 & b & 0 \\ 0 & 4 & c+a & -1 \end{bmatrix} \xrightarrow{\frac{R2}{2}} \begin{bmatrix} 1 & 3 & a & -1 \\ 0 & 1 & b/2 & 0 \\ 0 & 4 & c+a & -1 \end{bmatrix} \xrightarrow{R3-(4)R2} \begin{bmatrix} 1 & 3 & a & -1 \\ 0 & 1 & b/2 & 0 \\ 0 & 0 & c+a-2b & -1 \end{bmatrix}$$

$$\xrightarrow{R1-(3)R2} \begin{bmatrix} 1 & 0 & a-(3b/2) & -1 \\ 0 & 1 & b/2 & 0 \\ 0 & 0 & c+a-2b & -1 \end{bmatrix} \xrightarrow{\frac{R3}{c+a-2b}} \begin{bmatrix} 1 & 0 & a-(3b/2) & -1 \\ 0 & 1 & b/2 & 0 \\ 0 & 0 & 1 & -1/(a+c-2b) \end{bmatrix}$$

$$\xrightarrow{R2-(b/2)R3}\begin{bmatrix} 1 & 0 & a-(3b/2) & -1 \\ 0 & 1 & 0 & \dfrac{b}{2(a+c-2b)} \\ 0 & 0 & 1 & \dfrac{-1}{a+c-2b} \end{bmatrix}\xrightarrow{R1+[(3b/2)-a)]\times R3}\begin{bmatrix} 1 & 0 & 0 & -1+\dfrac{a-(3b/2)}{a+c-2b} \\ 0 & 1 & 0 & \dfrac{b}{2(a+c-2b)} \\ 0 & 0 & 1 & \dfrac{-1}{a+c-2b} \end{bmatrix}$$

The above matrix is in reduced row echelon form. Now,

$$\begin{bmatrix} 1 & 0 & 0 & -1+\dfrac{a-(3b/2)}{a+c-2b} \\ 0 & 1 & 0 & \dfrac{b}{2(a+c-2b)} \\ 0 & 0 & 1 & \dfrac{-1}{a+c-2b} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \dfrac{-1}{2} \\ 0 & 1 & 0 & \dfrac{-1}{6} \\ 0 & 0 & 1 & \dfrac{1}{6} \end{bmatrix}$$

$$\begin{aligned} -1+\frac{a-(3b/2)}{a+c-2b} &= \frac{-1}{2} \\ \frac{a-(3b/2)}{a+c-2b} &= \frac{1}{2} \\ a-(3b/2) &= a+c-2b \\ 2b-(3b/2) &= c \\ b/2 &= c \\ b &= 2c \qquad (1) \end{aligned}$$

$$\frac{b}{2(a+c-2b)} = \frac{-1}{6}$$

$$\begin{aligned} 6b &= 4b-2a-2c \\ a &= -b-c \qquad (2) \end{aligned}$$

$$\frac{-1}{a+c-2b} = \frac{1}{6}$$

$$-6 = a+c-2b \qquad (3)$$

Substitute (1) in (2)

$$\begin{aligned} a &= -2c-c \\ a &= -3c \qquad (4) \end{aligned}$$

Substitute $a = -3c$, $b = 2c$ in $a + b + c$

$$a + b + c = -3c + 2c + c$$
$$a + b + c = 0$$

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7. Question

Three mobile shops- shop A, shop B and shop C, sell three brands of mobile phones: brand R, brand S and brand T. In a week, shop A sold 1 mobile phone of brand R, 3k mobile phones of brand S, and 3k+4 mobile phones of brand T. Shop B sold 1 mobile phone of brand R, k+4 mobile phones of brand S, and 4k+2 mobile phones of brand T. Shop C sold 1 mobile phone of brand R, 2k+2 mobile phones of brand S, and 3k+4 mobile phones of brand T (assume, k≠2). Assume that the price of a given model of a given brand is the same in all the shops. Shop A, shop B, and shop C earned ₹61, ₹65 and ₹66 (in thousands), respectively by selling these three brands of mobile phones. If the price of each mobile phone of brand S is ₹5 (in thousands), then what is the price of each mobile phone of brand T (in thousands)? [Note: Suppose the price comes out to be 20,000, then the answer should be 20]?

7.1. Answer : 6

7.2. Solution

The following are the system of linear equations,

$$R + (3k)S + (3k+4)T = 61$$
$$R + (k+4)S + (4k+2)T = 65$$
$$R + (2k+2)S + (3k+4)T = 66$$

The matrix representation $Ax = b$ for the given system of linear equations is:

$$A = \begin{bmatrix} 1 & 3k & 3k+4 \\ 1 & k+4 & 4k+2 \\ 1 & 2k+2 & 3k+4 \end{bmatrix}, \; x = \begin{bmatrix} R \\ S \\ T \end{bmatrix}, \; b = \begin{bmatrix} 61 \\ 65 \\ 66 \end{bmatrix}$$

Augment A and b,

$$(A|b) = \left[\begin{array}{ccc|c} 1 & 3k & 3k+4 & 61 \\ 1 & k+4 & 4k+2 & 65 \\ 1 & 2k+2 & 3k+4 & 66 \end{array}\right]$$

Perform elementary row operations on $(A|b)$,

$$\left[\begin{array}{ccc|c} 1 & 3k & 3k+4 & 61 \\ 1 & k+4 & 4k+2 & 65 \\ 1 & 2k+2 & 3k+4 & 66 \end{array}\right] \xrightarrow{R2-R1} \left[\begin{array}{ccc|c} 1 & 3k & 3k+4 & 61 \\ 0 & 4-2k & k-2 & 4 \\ 1 & 2k+2 & 3k+4 & 66 \end{array}\right] \xrightarrow{R3-R1} \left[\begin{array}{ccc|c} 1 & 3k & 3k+4 & 61 \\ 0 & 4-2k & k-2 & 4 \\ 0 & 2-k & 0 & 5 \end{array}\right]$$

$$\xrightarrow{R2 \longleftrightarrow R3} \left[\begin{array}{ccc|c} 1 & 3k & 3k+4 & 61 \\ 0 & 2-k & 0 & 5 \\ 0 & 4-2k & k-2 & 4 \end{array}\right] \xrightarrow{R3-2(R2)} \left[\begin{array}{ccc|c} 1 & 3k & 3k+4 & 61 \\ 0 & 2-k & 0 & 5 \\ 0 & 0 & k-2 & -6 \end{array}\right]$$

$$\xrightarrow{\frac{R2}{2-k}} \left[\begin{array}{ccc|c} 1 & 3k & 3k+4 & 61 \\ 0 & 1 & 0 & \frac{5}{2-k} \\ 0 & 0 & k-2 & -6 \end{array}\right] \xrightarrow{\frac{R3}{k-2}} \left[\begin{array}{ccc|c} 1 & 3k & 3k+4 & 61 \\ 0 & 1 & 0 & \frac{5}{2-k} \\ 0 & 0 & 1 & \frac{-6}{k-2} \end{array}\right] \xrightarrow{R1-(3k)R2} \left[\begin{array}{ccc|c} 1 & 0 & 3k+4 & 61-\frac{15k}{2-k} \\ 0 & 1 & 0 & \frac{5}{2-k} \\ 0 & 0 & 1 & \frac{-6}{k-2} \end{array}\right]$$

$$\xrightarrow{R1-(3k+4)R3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 61-\frac{15k}{2-k}+\frac{18k+24}{k-2} \\ 0 & 1 & 0 & \frac{5}{2-k} \\ 0 & 0 & 1 & \frac{-6}{k-2} \end{array}\right]$$

The above matrix is in reduced row echelon form. From this, we get the following:

$$\begin{aligned} R &= 61-\frac{15k}{2-k}+\frac{18k+24}{k-2} \\ R &= 61-\frac{15k-18k-24}{2-k} \\ R &= 61-\left(\frac{3k+24}{2-k}\right) \qquad (1) \end{aligned}$$

$$S = \frac{5}{2-k} \qquad (2)$$

$$T = \frac{-6}{k-2} \qquad (3)$$

Given $\Longrightarrow k \neq 2$ and $S = 5$. From (2), we get:

$$\begin{aligned} 5 &= \frac{5}{2-k} \\ 2-k &= 1 \\ k &= 1 \end{aligned}$$

Substitute $k = 1$ in (1) and (3),

$$(1) \Longrightarrow R = \frac{61-(3k+24)}{2-k} = \frac{61-(27)}{1} = 34$$

$$(3) \Longrightarrow T = \frac{-6}{k-2} = \frac{-6}{1-2} = 6$$

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8. Question

The number of solutions of the system of equations is

$$\begin{aligned} 2x + 3y + 5z &= 1 \\ x + 2y + 3z &= 1 \\ x + y + 2z &= 25 \end{aligned}$$

8.1. Answer : 0

8.2. Solution

The matrix form ($Ax = b$) for the given system of linear equations is:

$$A = \begin{bmatrix} 2 & 3 & 5 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}, \; x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \; b = \begin{bmatrix} 1 \\ 1 \\ 25 \end{bmatrix}$$

Augment A and b,

$$(A|b) = \left[\begin{array}{ccc|c} 2 & 3 & 5 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 25 \end{array}\right]$$

Perform elementary row operations on $(A|b)$,

$$\left[\begin{array}{ccc|c} 2 & 3 & 5 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 25 \end{array}\right] \xrightarrow{\frac{R1}{2}} \left[\begin{array}{ccc|c} 1 & 3/2 & 5/2 & 1/3 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 25 \end{array}\right] \xrightarrow{R2-R1} \left[\begin{array}{ccc|c} 1 & 3/2 & 5/2 & 1/3 \\ 0 & 1/2 & 1/2 & 2/3 \\ 1 & 1 & 2 & 25 \end{array}\right]$$

$$\xrightarrow{R3-R1} \left[\begin{array}{ccc|c} 1 & 3/2 & 5/2 & 1/3 \\ 0 & 1/2 & 1/2 & 2/3 \\ 0 & -1/2 & -1/2 & 74/3 \end{array}\right] \xrightarrow{R3+R2} \left[\begin{array}{ccc|c} 1 & 3/2 & 5/2 & 1/3 \\ 0 & 1/2 & 1/2 & 2/3 \\ 0 & 0 & 0 & 76/3 \end{array}\right] \xrightarrow{R2\times 2} \left[\begin{array}{ccc|c} 1 & 3/2 & 5/2 & 1/3 \\ 0 & 1 & 1 & 4/3 \\ 0 & 0 & 0 & 76/3 \end{array}\right]$$

$$\xrightarrow{R1-(3/2)R2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -5/3 \\ 0 & 1 & 1 & 4/3 \\ 0 & 0 & 0 & 76/3 \end{array}\right] \xrightarrow{R3\times(3/76)} \left[\begin{array}{ccc|c} 1 & 0 & 1 & -5/3 \\ 0 & 1 & 1 & 4/3 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

The above matrix is in reduced row echelon form. When converting the third row into a linear equation, we get:

$$0x + 0y + 0z = 1$$

This is not possible. Hence, there is no solution for the given system of linear equations.

---

9. Question

A is the reduced row echelon form of the matrix $\begin{bmatrix} 1 & 3 & 0 & 0 \\ 4 & 1 & 5 & 24 \\ 2 & 2 & 7 & 59 \\ 3 & 9 & 0 & 0 \end{bmatrix}$. Then determinant of A is

9.1. Answer : 0

9.2. Solution

Perform elementary row operations on A,

$$\begin{bmatrix} 1 & 3 & 0 & 0 \\ 4 & 1 & 5 & 24 \\ 2 & 2 & 7 & 59 \\ 3 & 9 & 0 & 0 \end{bmatrix} \xrightarrow{R2-(4)R1} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & -11 & 5 & 24 \\ 2 & 2 & 7 & 59 \\ 3 & 9 & 0 & 0 \end{bmatrix} \xrightarrow{R3-(2)R1} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & -11 & 5 & 24 \\ 0 & -4 & 7 & 59 \\ 3 & 9 & 0 & 0 \end{bmatrix} \xrightarrow{R4-(3)R1} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & -11 & 5 & 24 \\ 0 & -4 & 7 & 59 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow[-11]{\frac{R2}{}} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & -5/11 & -24/11 \\ 0 & -4 & 7 & 59 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R3+(4)R2} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & -5/11 & -24/11 \\ 0 & 0 & 57/11 & -37/11 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R3\times\frac{11}{57}} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & -5/11 & \frac{-24}{11} \\ 0 & 0 & 1 & \frac{-37}{11} \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R2+\left(\frac{5}{11}\right)R3} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 0 & \frac{-24}{11}-\frac{185}{627} \\ 0 & 0 & 1 & \frac{-37}{11} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R1-(3)R2} \begin{bmatrix} 1 & 0 & 0 & 3\left(\frac{24}{11}+\frac{185}{627}\right) \\ 0 & 1 & 0 & \frac{-24}{11}-\frac{185}{627} \\ 0 & 0 & 1 & \frac{-37}{11} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The above matrix is in reduced row echelon form. All the elements of the last row are zero. Hence, $det(A) = 0$.

Notice the matrix after the 3rd row operation, all the elements in the last row became zero. We could have concluded that the determinant of the reduced row echelon form of the given matrix is zero after the third row operation itself.

---

10. Question

If $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is a solution of the system of equations, then the value of $x + y + z$ is

$$7x + 2y + z = 1$$

$$
\begin{aligned}
3y - z &= 1 \\
-3x + 4y - 2z &= 2
\end{aligned}
$$

10.1. Answer : 19

10.2. Solution

- The matrix form ($Ax = b$) for the given system of linear equations is:

$$A = \begin{bmatrix} 7 & 2 & 1 \\ 0 & 3 & -1 \\ -3 & 4 & -2 \end{bmatrix}, \; x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \; b = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

- Calculate determinant of A.

$$
\begin{aligned}
det(A) &= 7 \times (-6 - (-4)) - 3 \times (-2 - (3)) \\
det(A) &= 1
\end{aligned}
$$

- $det(A) \neq 0$. Therefore, the system of linear equations have unique solution. Hence, we shall proceed to find the solution of $x, y \; and \; z$ using Cramer's rule.

- Replace column 1 of A with b to get A_x,

$$A_x = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 3 & -1 \\ 2 & 4 & -2 \end{bmatrix}. \quad \begin{aligned} det(A_x) &= 1 \times (-6 - (-4)) - 1 \times (-4 - (4)) + 2 \times (-2 - (3)) \\ det(A_x) &= -4 \end{aligned}$$

- Replace column 2 of A with b to get A_y,

$$A_y = \begin{bmatrix} 7 & 1 & 1 \\ 0 & 1 & -1 \\ -3 & 2 & -2 \end{bmatrix}. \quad \begin{aligned} det(A_y) &= 7 \times (-2 - (-2)) - 3 \times (-1 - (1)) \\ det(A_y) &= 6 \end{aligned}$$

- Replace column 3 of A with b to get A_z,

$$A_z = \begin{bmatrix} 7 & 2 & 1 \\ 0 & 3 & 1 \\ -3 & 4 & 2 \end{bmatrix}. \quad \begin{aligned} det(A_z) &= 7 \times (6 - (4)) - 3 \times (2 - (3)) \\ det(A_z) &= 17 \end{aligned}$$

- By Cramer's rule,

$$x = \frac{det(A_x)}{det(A)} = \frac{-4}{1} = -4$$

$$y = \frac{det(A_y)}{det(A)} = \frac{6}{1} = 6$$

$$x = \frac{det(A_z)}{det(A)} = \frac{17}{1} = 17$$

- Find $x+y+z$,

$$\begin{aligned} x+y+z &= -4+6+17 \\ x+y+z &= 19 \end{aligned}$$

---

11. Question

Let $A = \begin{bmatrix} 2 & 10 & 2 & 8 \end{bmatrix}$ and M denote the reduced row echelon form of $A^T A$. The number of non-zero
rows of M is

11.1. Answer: 1

11.2. Soluton

Find $A^T A$,

$$A^T A = \begin{bmatrix} 4 & 20 & 4 & 16 \\ 20 & 100 & 20 & 80 \\ 4 & 20 & 4 & 16 \\ 16 & 80 & 16 & 64 \end{bmatrix}$$

Apply elementary row operations on $A^T A$,

$$\begin{bmatrix} 4 & 20 & 4 & 16 \\ 20 & 100 & 20 & 80 \\ 4 & 20 & 4 & 16 \\ 16 & 80 & 16 & 64 \end{bmatrix} \xrightarrow{\frac{R1}{4}} \begin{bmatrix} 1 & 5 & 1 & 4 \\ 20 & 100 & 20 & 80 \\ 4 & 20 & 4 & 16 \\ 16 & 80 & 16 & 64 \end{bmatrix} \xrightarrow{R2-(20)R1} \begin{bmatrix} 1 & 5 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 4 & 20 & 4 & 16 \\ 16 & 80 & 16 & 64 \end{bmatrix}$$

$$\xrightarrow{R3-4(R1)} \begin{bmatrix} 1 & 5 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 16 & 80 & 16 & 64 \end{bmatrix} \xrightarrow{R4-(16)R1} \begin{bmatrix} 1 & 5 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now, the above matrix is in reduced row echelon form. As we can see, there is only one non-zero row in M.

---

12. Question

Consider the curve corresponding to the function $f(x) = ax^3 + bx^2 + cx + d$. The following points lie on the curve: $(1,1), (2,7), (-1,-5), (3,23)$. Find the value of a−b+c−d.

12.1. Answer : 5

12.2. Solution

Substitue the $x - coordinate$ of $(-1,-5)$ in $f(x)$ and equate it with the $y - coordinate$.

$$\begin{aligned} a(-1)^3 + b(-1)^2 + c(-1) + d &= -5 \\ -a + b - c + d &= -5 \end{aligned}$$

$$a - b + c - d = 5$$

Comprehension $(13, 14, 15)$

The network in $Figure: \ M2W2GA1$ shows a proposed plan for flow of traffic around a park. All the streets are assumed to be one-way and the arrows denote the direction of flow of traffic. The plan calls for a computerized traffic light at the South Street. Let $2x_1, 3x_2, 2x_3 \ and \ x_4$ denote the average number (per hour) of vehicles expected to pass through the connecting streets (e.g., $2x_1$ denote the average number (per hour) of vehicles expected to pass through the street connecting the North Street and West Street as shown in Figure: $M2W2GA1$). 400, 1000, 900, and c denote the average number (per hour) of vehicles expected to pass through West, North, East, and South Streets respectively.

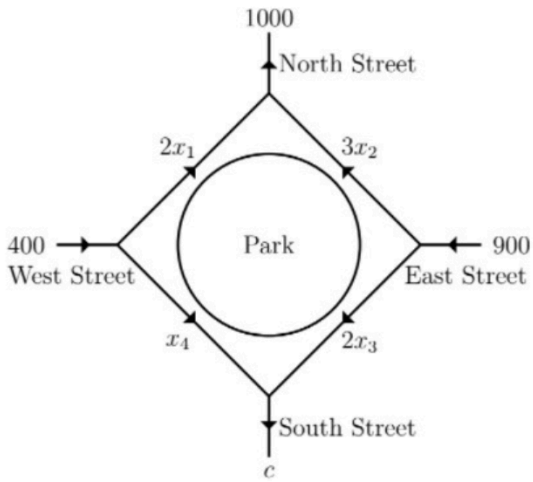

Figure: M2W2GA1

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13. Question

Which of the following options are correct?

The system of equations corresponding to the flow of expected traffic according to the given data above, will be

☐
$$\begin{aligned} 2x_1 + 3x_2 &= 1000 \\ 3x_2 + 2x_3 &= 900 \\ 2x_3 + x_4 &= c \\ 2x_1 + x_4 &= 400 \end{aligned}$$

The system of equations corresponding to the flow of expected traffic according to the given data above, will be

☐
$$\begin{aligned} 2x_1 + 3x_2 &= 900 \\ 3x_2 + 2x_3 &= 1000 \\ 2x_3 + x_4 &= 400 \\ 2x_1 + x_4 &= c \end{aligned}$$

The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is

- [ ] $$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 900 \\ 1000 \\ 400 \\ c \end{bmatrix}$$

The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is

- [ ] $$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1000 \\ 900 \\ c \\ 400 \end{bmatrix}$$

13.1. Answers :

- The system of equations corresponding to the flow of expected traffic according to the given data above, will be

$$\begin{aligned} 2x_1 + 3x_2 &= 1000 \\ 3x_2 + 2x_3 &= 900 \\ 2x_3 + x_4 &= c \\ 2x_1 + x_4 &= 400 \end{aligned}$$

- The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is

$$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1000 \\ 900 \\ c \\ 400 \end{bmatrix}$$

13.2. Solution

- On an average, 1000 vehicles pass through the North Street. Of those, $2x_1$ vehicles are from the East Street and $3x_2$ vehicles are from the West Street. Therefore, $2x_1 + 3x_2 = 1000$.
- On an average, 900 vehicles pass through the East street. Of those, $3x_2$ vehicles travel toward the North street and $2x_3$ vehicles trave towards South Street. Therefore, $3x_2 + 2x_3 = 900$.
- On an average, c vehicles pass through the South Street. Of those, x_4 vehicles are from the West Street and $2x_3$ vehicles are from the East Street. Therefore, $x_4 + 2x_3 = c$.
- On an average, 400 vehicles pass through the West street. Of those, $2x_1$ vehicles travel towards the North Steet and x_4 vehicles travel towards South Street. Therefore, $2x_1 + x_4 = 400$.
- From the above four points, we can see that **option 1** is correct.

The matrix representation $Ax = b$ for the above four systm of linear equations is:

$$A = \begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix}, \; x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, \; b = \begin{bmatrix} 1000 \\ 900 \\ c \\ 400 \end{bmatrix}$$

Hence, **option 4** is correct.

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14. Question

How many vehicles are expected to pass through the South Street per hour on an average?

14.1. Answer : 300

14.2. Solution

$$\begin{aligned} 2x_1 + 3x_2 &= 1000 \quad (1) \\ 3x_2 + 2x_3 &= 900 \quad (2) \\ 2x_3 + x_4 &= c \quad (3) \\ 2x_1 + x_4 &= 400 \quad (4) \end{aligned}$$

By $(1) - (2)$,

$$\begin{aligned} 2x_1 - 2x_3 &= 100 \\ x1 - x3 &= 50 \quad (5) \end{aligned}$$

By $(4) - (3)$,

$$\begin{aligned} 2x_1 - 2x_3 &= 400 - c \\ x_1 - x_3 &= 200 - (c/2) \quad (6) \end{aligned}$$

By $(6) - (5)$,

$$\begin{aligned} (x_1 - x_3) - (x_1 - x_3) &= 200 - (c/2) - 50 \\ (c/2) &= 150 \\ c &= 300 \end{aligned}$$

Therefore, on an average, 300 vehicles are expected to pass through the South Street per hour.

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15. Question

Match the names of the street in Column A with the maximum and minimum number of vehicles expected to pass through the street on an average (per hour) in Column B and Column C, respectively; in $Table\ M2W2GA1$.

	Name of the connecting street		The maximum number of vehicles expected to pass through the street (per hour)		The minimum number of vehicles expected to pass through the street (per hour)
	Column A		Column B		Column C
a)	Connecting West and North street	i)	300	1)	0
b)	Connecting East and North street	ii)	300	2)	100
c)	Connecting East and South street	iii)	400	3)	600
d)	Connecting West and South street	iv)	900	4)	0

Table : M2W2GA1

- [ ] d → i→ 2
- [ ] b→ iii → 3.
- [ ] b→ iv→ 3.
- [ ] d→ ii → 1.
- [ ] a→ iii→ 2.
- [ ] a→ iii→ 4.
- [ ] c→ i→ 4.

15.1. Answers :

- b→ iv→ 3.
- d→ ii → 1.
- a→ iii→ 2.
- c→ i→ 4.

15.2. Solution

- From the West Street, if all the 400 vehicles pass towards the North Street, then no vehicle would pass towards the South Street.
- From the West Street, the maximum number of vehicles which can pass towards the South Street is 300. In that case, the minimum number of vehicles which would pass towards the North Street is 100.
- From the East Street, if all the 900 vehicles pass towards the North Street, then no vehicles would pass towards the South Street.
- From the East Street, the maximum number of vehicles which can pass towards the South Street is 300. In that case, the minimum number of vehicles which would pass towards the North Street is 600.
- From East Street, if 600 vehicles pass toward the North Street, then we can expect 400 vehicles to pass from West Street towards North Street. In this case, we can expect 300 vehicles to pass from East Street towards South Street and no vehicles to pass from the West Street towards the South Street.
- From West Street, if 300 vehicles pass toward the South Street, then we can expect no vehicle to pass from East Street towards South Street. In this case, we can expect 100 vehicles to pass from West Street towards North Street and 900 vehicles to pass from the East Street towards the North Street.

From the above observations, we can see that **option 3, option 4, option 5 and option 7** are correct.